Understanding statistical tests of homogeneity

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Abstract

This report investigates the statistical tests for homogeneity of distributions that relies on the chi2-squared distribution. We generate samples from the multinomial distribution by computing histograms from generated samples of binomial random variables. We then investigate how the probability of a false negative in the hypothesis test is affected by the strength of the signal, the amount of data that is used to construct the contigency table and the number of categories in our samples in the multinomial distributions that are being sampled.

1 Introduction

A statistical test of homogeneity starts by supposing that a set of n individuals have been classified in two ways. Each individual is within one of r mutually exclusive classes $A_1, A_2, \dots A_r$ **and** one of c mutually exclusive classes $B_1, B_2, \dots B_r$. We can then write out the number of individuals, O_{ij}, in each of the $r \times c$ resulting categories in a contingency table:

	B_1	$\dots$	B_j	$\dots$	B_c	Row sum
A_1	O_{11}	$\dots$	O_{1j}	$\dots$	O_{1c}	$\sum_{j=1}^c O_{1j}$
$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
A_i	O_{i1}	$\dots$	O_{ij}	$\dots$	O_{ic}	$\sum_{j=1}^c O_{ij}$
$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
A_r	O_{r1}	$\dots$	O_{rj}	$\dots$	O_{rc}	$\sum_{j=1}^c O_{rj}$
Column sum	$\sum_{i=1}^r O_{i1}$	$\dots$	$\sum_{i=1}^r O_{ij}$	$\dots$	$\sum_{i=1}^r O_{ic}$	n $= \sum_{i=1}^r \sum_{j=1}^c O_{ij}$

A test for homogeneity tests the null hypothesis $P(A_i \cap B_j) = P(A_i)P(B_j)$ against the alternative that $P(A_i \cap B_j) \neq P(A_i)P(B_j)$*. The test statistic for this test is computed as:

$$\phi^2 = \sum_{1=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \qquad \text{where} \qquad E_{ij} = \frac{\left(\sum_{k=1}^c O_{ik}\right)\left(\sum_{l=1}^r O_{lj}\right)}{n} \tag{1}$$

*Note that this is the way the null and alternative hypothesis are setup for the related test of independence of classifications. The null and alternative hypothesis for a test of homogeneity are difference because, when you are comparing two distributions, you set all the $P(A_i)$ values when you choose how many samples to generate from each distribution

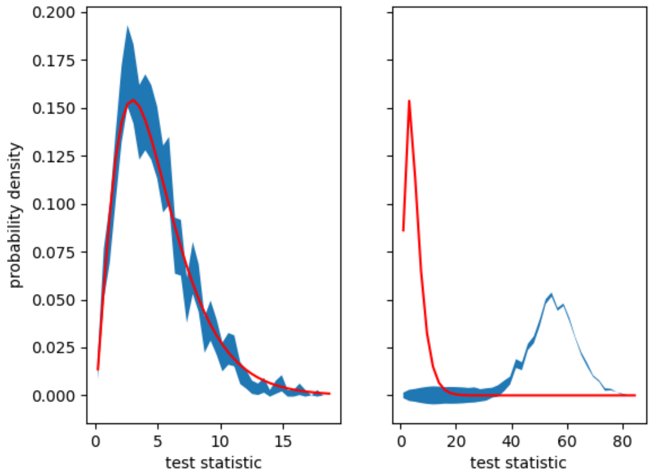

Fig. 1. The blue shaded regions show the estimates of the histograms for the test statistic that was defined in equation 1. To generate each of the 2000 instances of the test statistic that were used to construct the histogram with 40 bins we generated two samples of 50 binomial random variables. The first of these was for a binomial random variable with $n = 5$ and $p = p_1$, while the second was for a binomial random variable with $n = 5$ and $p = p_2$. In the left panel $p_1 = p_2 = 0.4$ while in the right $p_1 = 0.4$ and $p_2 = 0.8$. The shaded region shows the 90% confidence limit on the estimate of the histogram that was obtained by sampling. The red lines are then the probability density function for a chi-squared distribution with five degrees of freedom.

In the following sections we will investigate the distribution that is sampled by this statistic by constructing multiple histograms containing samples from binomial random variables.

2 Results and discussion

A sample of the test statistic defined by equation 1 can be generated by:

1. Generating 50 binomial random variables with $n = 5$ and $p = p_1$.
2. Generating 50 binomial random variables with $n = 5$ and $p = p_2$.
3. Computing two separate histograms with $n + 1$ bins from these two samples.
4. Constructing an $(n + 1) \times 2$ contingency table with these two histograms.
5. Computing the test statistic defined by equation 1 using our contingency table.

Figure 1 shows the results that we obtained when we performed this set of operations 2000 times and constructed a histogram with 40 bins from the resulting samples. For the histogram in the left panel we set $p_1 = p_2 = 0.4$, while for the right panel we set $p_1 = 0.4$ and $p_2 = 0.8$.

The red lines in figure 1 show the probability density function for a chi-squared distribution with 5 degrees of freedom. You can see that when $p_1 = p_2$ there is insufficient evidence to say

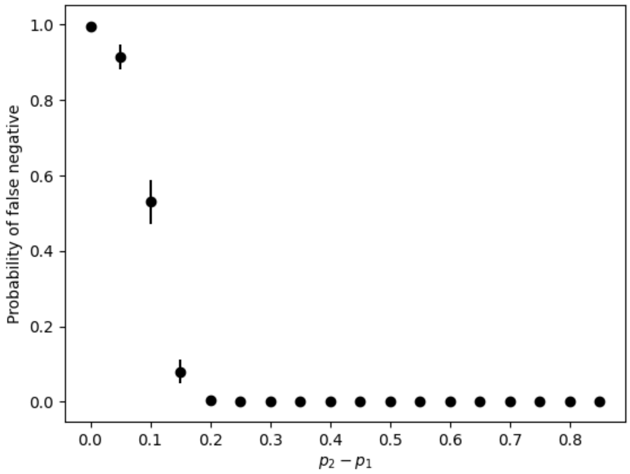

Fig. 2. Probability for a false negative result for a false negative result on a hypothesis test at the 5% significance level in which equation 2 is used to determine the p-value as a function of the difference between the p parameters of the two binomial random variables whose histograms were compared. Each black dot is the gives the fraction of 200 hypothesis tests that resulted in a p-value that was greater than 0.05. Each of these hypothesis tests compared the histograms that were constructed from samples of two binomial random variables with $n = 5$ and $p = 0.1$ and $n = 5$ and $p = p_2 - p_1 + 0.1$ respectively. The error bars show the 90% confidence limit on our estimate of the false negative rate.

that the statistic defined by equation 1 is not a sample from this distribution[†]. However, for the histogram that was constructed with $p_1 \neq p_2$ the statistic we are computing is clearly not a sample from this distribution.

Knowing that, under the null hypothesis, the test statistic defined by equation 1 is a sample from a chi-squared distribution with $(r-1)(c-1)$ degrees of freedom allows us to calculate a p-value for a statistical test of the null hypothesis $P(A_i \cap B_j) = P(A_i)P(B_j)$ against the alternative that $P(A_i \cap B_j) \neq P(A_i)P(B_j)$ using:

$$p = 1 - \chi_{(r-1)(c-1)}(\phi^2) \tag{2}$$

where $\chi_{(r-1)(c-1)}$ is the cumulative distribution for the chi-squared distribution with $(r-1)(c-1)$ degrees of freedom.

Figure 2 shows how frequently we get a false negative result from a hypothesis test using this expression for the p-value and a 5% significance level for different values of $p_2 - p_1$. To generate each of the points in this figure we performed 200 hypothesis tests. For each of these tests we constructing a first histogram from a sample of 50 binomial random variables with $p = 0.1$ and $n = 5$. A second histogram was then constructed from a sample of 50 binomial random variables with $p = p_2 - p_1 + 0.1$ and $n = 5$. The points in figure 2 indicate the fraction of times the p-value evaluated using equation 2 was greater than 0.05. You can

[†]To be clear, the statistic is a sample from this distribution but we cannot prove that it is using sampling

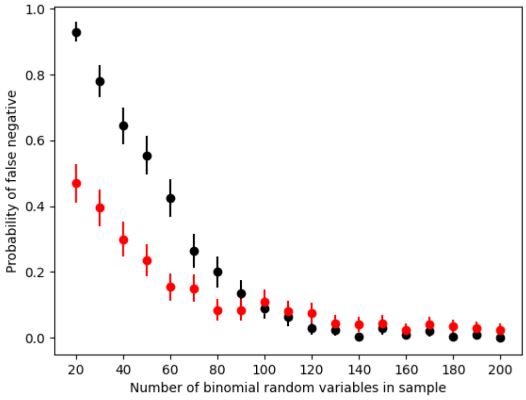

Fig. 3. Probability for a false negative result on a hypothesis test at the 5% significance level in which equation 2 is used to determine the p-value as a function of the number of binomial random variables that were used to construct histograms. Each dot gives the fraction of 200 hypothesis tests that resulted in a p-value that was greater than 0.05. Each of these hypothesis tests compared the histograms that were constructed from samples of two binomial random variables with $n = 5$ and $p = 0.1$ and $n = 5$ and $p = 0.2$ respectively. The error bars show the 90% confidence limit on our estimate of the false negative rate. The black dots show the results that were obtained when two samples of binomial random variables had the size indicated on the x-axis. The red dots show the results that were obtained when only 100 samples of the binomial random variable with $p = 0.1$ were generated and the x-axis gives the number of samples of the binomial random variable with $p = 0.2$.

see that false positives are extremely likely when $p_2 - p_1$ is small but also that false positives are extremely unlikely once $p_2 - p_1 > 0.2$.

It seems likely that if more binomial random variables are used to construct the histograms the statistical test will be able to recognize even small differences between the p_1 and p_2 parameters of the random variables. Figure 3 confirms that this is the case. The black points in this figure were once again constructed by performing 200 hypothesis tests. For each of these tests we compared histograms constructed from two binomial random variables with $n_1 = 5$ and $p_1 = 0.1$ and $n_2 = 5$ and $p_2 = 0.2$, respectively. The x axis in this figure indicates the number of random variables in the two samples that were used to construct the histograms that were then used when evaluating equation 2.

The red points in figure 3 indicate that increasing the amount of data in only one of the samples that are used when computing equation 2 also reduces the likelihood of false negative results. These values were obtained like the black points by performing 200 hypothesis tests and by generating binomial random variables with $n_1 = 5$ and $p_1 = 0.1$ and $n_2 = 5$ and $p_2 = 0.2$, respectively. The difference between the red points and the black points is that the x axis shows how many binomial random variables with $p_2 = 0.2$ were generated. The histograms for the binomial random variable with $p_1 = 0.1$ were always constructed from a sample of 100 random variables.

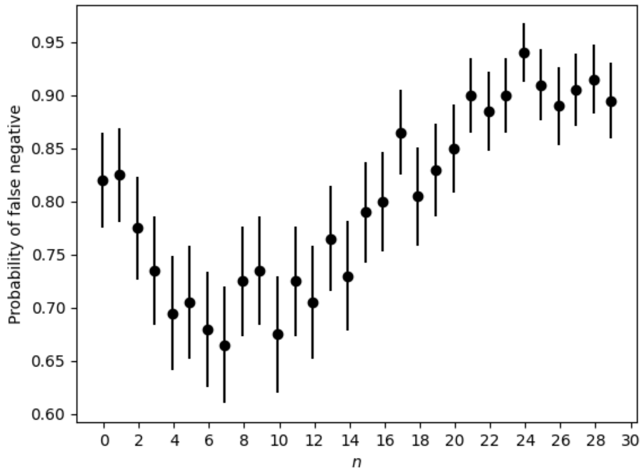

Fig. 4. Probability for a false negative result on a hypothesis test at the 5% significance level in which equation 2 is used to determine the p-value as a function of the n parameters that were used to construct the histograms. Each dot gives the fraction of 200 hypothesis tests that resulted in a p-value that was greater than 0.05. Each of these hypothesis tests compared the histograms that were constructed from samples of two binomial random variables with $n = x$ and $p = 0.4$ and $n = x$ and $p = 0.5$ respectively. The error bars show the 90% confidence limit on our estimate of the false negative rate.

Figure 4 shows that if there are more categories for your distributions then more data is required to recognize differences between the various histograms that are generated. To generate each of the points in this figure we performed 200 hypothesis tests. For each of these tests we constructing a first histogram from a sample of 50 binomial random variables with $p = 0.4$ and $n = x$. A second histogram was then constructed from a sample of 50 binomial random variables with $p = 0.5$ and $n = x$. The points in figure 4 indicate the fraction of times the p-value that we get by constructing a contingency table from these two histograms and using it to evaluate equation 2 was greater than 0.05. Figure 4 shows that the probability of a false negative is larger when the n parameter of the generated binomial random variables is both small and large. The increase that is seen for large values of n is due to the limited amount of data. The proportion of false negatives for low n is likely a consequence of the fact that difference between the probability mass functions for a binomial random variables with parameters n and $p = 0.4$ and parameters n and $p = 0.5$ decreases with n.